

Using URL Characteristics to Detect Phishing Websites Through Logistic Regression

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Abstract—This paper analyzes the UCI Phishing Websites dataset and develops a logistic regression model to classify URLs as phishing or legitimate. The model achieves 92.8% accuracy and an AUC of 0.98. Analysis of logistic regression coefficients shows that features such as URL_of_Anchor, SSLfinal_State, Prefix_Suffix, web_traffic, and SFH are the strongest predictors of phishing. Results demonstrate that interpretable machine learning can effectively identify risky URL characteristics.

Index Terms—phishing, machine learning, cybersecurity, logistic regression, URL analysis

I. INTRODUCTION

Phishing attacks still rank among the top online dangers [1], [2]. Instead of guessing, this work checks what parts of web addresses give clues about scams, using a method called logistic regression to spot risky pages. That technique relies on number patterns and shows up often here. It reviews different address details, then decides, based on data - if a page seems trustworthy.

The core goal here is building fast tools that stop fake websites right away - keeping harm from happening. Instead of waiting, systems could spot bad signs in web addresses early. This means safer browsing without slowing things down. Focusing on how scams hide in links makes defenses sharper over time [3].

II. RELATED WORK

Past work on spotting fake websites found that using smart algorithms works well when telling bad links from real ones [4]–[8]. Research shows tools like tree models, line separators, brain-like nets, or mixed systems can score solid results - sometimes even above 95 percent right calls. Instead of one single clue, they usually mix link patterns, page text, and server traits to catch scams. Even though their guesses hit the mark often, understanding exactly why they flag a site isn't always clear.

Some experts point out how crucial it is to understand phishing detectors, especially when used in actual security setups [3], [9]. Knowing why a link gets marked as malicious builds user confidence, aids analysts during choices, while guiding firms toward stronger protection methods. Still, plenty of past work focuses more on boosting precision than clarifying how models actually decide things. On top of that, many papers depend on data sets failing to capture today's shifting phishing trends, since attack styles keep transforming fast in both layout and tactics [2].

This project takes earlier research further - shifting focus to logistic regression, a model that's easier to interpret statistically, looking at phishing clues found just in URLs. Instead of chasing high accuracy alone, it uses coefficients to show which features really matter, tackling the trade-off between getting results and understanding them [1], [9]. Not just flagging bad links, but pinpointing what traits stand out most when classifying them, revealing patterns behind fake versus real websites.

III. DATASET DESCRIPTION

Our data comes from the UC Irvine Machine Learning Repository [10]. It was created by Rami Mohammad and Lee McCluskey in 2012 [11]. In the dataset, there are a total of 11,055 URLs, which are listed as "instances." Within these, 30 URL-based features were identified. Some of these features, notably the important ones, can be found within Table 1. A few examples include PrefixSuffix, webtraffic, GoogleIndex, and DNSRecord.

Within the dataset, the response variable lists these URLs as either phishing or legitimate. The result for these variables, are shown as either a "-1" or a "1." A negative value indicates that a URL was suspected to be a phishing attempt. A positive value indicates that the source was not a phishing attempt. The "variables table" lists each feature with a corresponding role,

type, description, unit, and potential missing values. However, according to the documentation, there were no missing values found, so all instances of this variable are listed as “no.”

IV. METHODOLOGY

The data was implemented entirely into Python. Python was utilized to access the scikit-learn library. The scikit-learn library was used for the model’s developmental process. Python was utilized to access pandas which were used for data management. The dataset came from UC Irvine’s Machine Learning Repository which was stored in an Attribute-RelationFile Format (ARFF) [10]. Arff and pandas libraries contained the data which was 11,055 URL’s and 30 URL-Based Features. Before processing the data, the data had to be cleaned. After the data was cleaned, feature scaling was enforced to normalize the data range. Features with a higher numerical value would not skew the data while the model processed the data.

Logistic Regression models were utilized to see which features of a URL had the biggest influence on being an illegitimate URL. The model will balance weights by default, so it can give equal effort into determining whether data was legitimate or phishing. Stratified sampling designated the data in an 80/20 split. 80 percent of the data was used for training the model while 20 percent was used to confirm results with final testing. This helps the model learn how to process new/unseen URLs.

V. RESULTS

Logistic regression proved to be more than effective when distinguishing between fake and real URLs. With a training-to-testing ratio of 80 to 20, it reached 92.8 percent accuracy, properly labeling 2054 out of 2211 web addresses during evaluation. Supporting evidence comes from the confusion matrix, which highlights strong detection of malicious links. At the same time, only a small portion of genuine sites were wrongly flagged. Overall, findings indicate this method can be trusted for spotting phishing efforts.

Beyond correctness, performance was checked via an ROC graph. With an AUC reaching 0.98, clear distinction appears between fake and real links. Since this score nears 1.0, phishing addresses tend to get higher likelihoods compared to genuine ones. That result supports how well logistic regression works here.

Looking at the model’s coefficients showed what parts of a URL affected phishing detection most. SSLfinalState, URLofAnchor, PrefixSuffix, webtraffic, along with SFH stood out because their values were largest in size. Because these factors carried more weight, they guided how the model sorted results. Earlier studies have found similar patterns - this supports those conclusions again here [4], [5], [9]. Specific traits in URLs matter greatly when spotting fake sites. Highlighting them helps explain why certain links get flagged, making outcomes easier to understand.

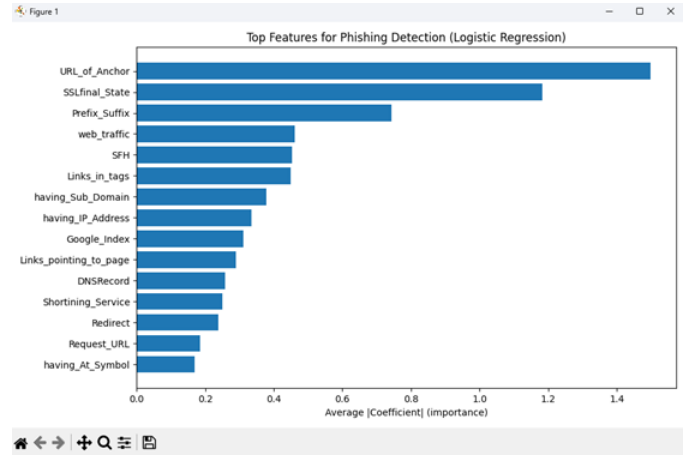


Fig. 1. Top features for phishing detection (logistic regression).

TABLE I
TOP 15 FEATURES BY IMPORTANCE (LOGISTIC REGRESSION)

Feature	Importance
URL_of_Anchor	1.498506
SSLfinal_State	1.182925
Prefix_Suffix	0.743102
web_traffic	0.460955
SFH	0.453332
Links_in_tags	0.448419
having_Sub_Domain	0.377851
having_IP_Address	0.335278
Google_Index	0.311220
Links_pointing_to_page	0.289072
DNSRecord	0.256945
Shortning_Service	0.250803
Redirect	0.237606
Request_URL	0.185940
having_At_Symbol	0.169147

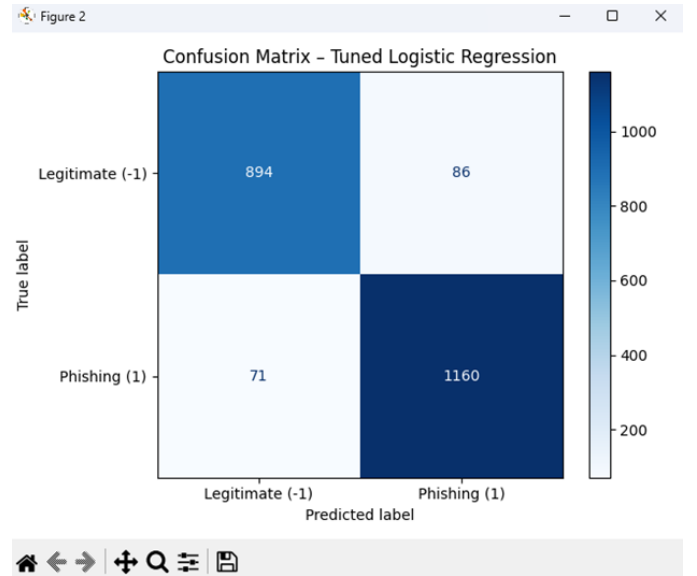


Fig. 2. Confusion matrix – tuned logistic regression.

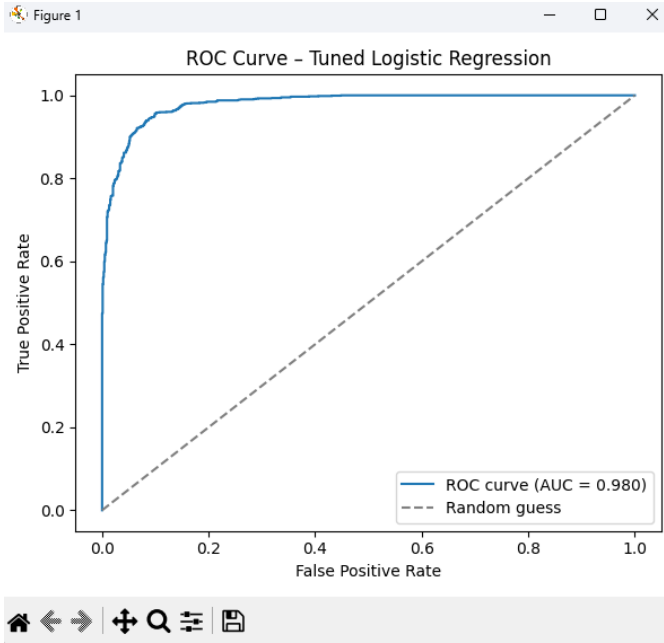


Fig. 3. ROC curve – tuned logistic regression (AUC = 0.980).

VI. DISCUSSION

The model was used to achieve an accuracy of 92.8 percent. The ROC-AUC score was 0.98. With these scores, it is safe to assume that the model can predict whether an URL is legitimate or a phishing URL. After observing the coefficients of the model, ‘red flags’ were able to be analyzed and indicated phishing components. Two of the most outlying and highest weighted indicators were “URL of Anchor” and “SSL Final State”. This confirms that the strongest indicator of a phishing URL is the discrepancies of visible text and where the link destination is, and also invalid security certificates.

The results help Cybersecurity professionals focus on what features of a URL are most susceptible to phishing attempts. The model uses interpretable rules which can be applied into many browsers and training programs. These tools will be used to train users on what feature patterns are present and what features of phishing URLs appear as. Users can directly use the model to determine whether a link is legitimate or not. The model’s highest weighted red flags can be assessed by Cybersecurity professionals. A Cybersecurity professional can view the red flag URL features on URLs they deem suspicious if they need to verify whether or not a URL is of phishing nature. It is important to remember that the study we concluded used a dataset based on features defined in 2012. Future research will need to assess how these features will be used to defend against ‘zero day’ attacks [6]. These zero-day attacks will most likely use techniques that are not present in this dataset.

VII. LIMITATIONS

The dataset used in the research had only included examples that were previously labeled as either phishing or legitimate,

meaning every entry presented in the dataset had been examined and categorized [10], [11]. The clear labeling was deemed helpful, as it described a trend to view. However, research limitations had begun to surface. As the data set had only included previously labeled examples, it did not fully represent the range of phishing risks. Focusing only on the labeled data, the models may miss any valuable information provided from the unlabeled data.

The logistical regression model only assumes the simple, linear relationship between URL features and phishing risks. This could lead to missing intricate patterns that are currently being presented in real phishing URLs. By only considering a simple linear relationship, this could lead to important details being missed that could further the research process. As stated previously, not all modern phishing techniques were documented in the research. It’s crucial to understand the newer techniques/features being used for damage to users. Without the proper knowledge of these new techniques, this leads to a decrease in predictiveness, as these techniques were not included.

VIII. CONCLUSION

In this report, we detailed our implementation of a logistical regression model for a phishing website dataset from the UC Irvine Machine Learning Repository [10]. We used a Python script to train a model that would eventually test itself by learning patterns about phishing URL differences discovered within the dataset. It was able to identify the most predictive phishing URL features with a 92.8 percent accuracy, and it produced a top-left leaning ROC curve near the 1.0 mark. With this information, we as a group consider this to be a successful model.

Based on our results, URLofAnchor and SSLfinalState were the top two biggest indicators of a phishing attempt. URLofAnchor and SSLfinalState were the only two features that scored above 1.0 on the scale of importance. The third most prominent feature, “PrefixSuffix,” scored 0.743102. This difference makes it evident that the two features were significantly more common than the other ones in the dataset. In practicality, these two variables deal with a website’s anchor tags and SSL certificates, respectively. This means that most detected phishing URLs had deceptive anchor tags or unverified SSL certificates.

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